

An Integrated Rigid-Flexible Body Dynamic Approach to Computationally Efficient Musculoskeletal Modeling and Muscle Recruitment Simulation of the Lumbosacral Spine and Torso

Siril Teja Dukupati & Mark Driscoll

Abstract— Objective: *In silico* biomechanical models of the spine traditionally follow either rigid body dynamic (RBD) modeling (multibody modeling) or finite element (FE) modeling techniques. While RBD models lack robust representation for flexible tissues, FE models are computationally expensive. This study proposes an integrated rigid-flexible body dynamic (RFBD) architecture to address these limitations, and develops a full-torso human model, focusing spinal mechanical stability. **Methods:** The model consisted of L1-L5 lumbar vertebrae, pelvis, sacrum, a lumped thoracic spine with ribcage as rigid bodies, while the intervertebral discs (IVDs), abdominal cavity and thoracolumbar fascia (TLF) were modeled as deformable reduced-order flexible bodies. Spinal ligaments were represented as nonlinear tension-only springs, while the musculature was modeled as tension-only forces. Level-by-level spinal stiffness was validated under pure flexion moments up to 7.5 Nm against literature studies. The reduced-order implementation was also validated against an identical FE model. Spinal stability contribution of different tissues in flexion was systematically evaluated using six on-off cases. **Results:** Passive spine segmental stiffness profiles matched well with *ex vivo* and *in silico* comparators. The RFBD method demonstrated strong agreement with the FE solver, while significantly reducing computational demand. Stability analyses highlighted the role of intra-abdominal pressure in spinal unloading and generation of compressive loads along the spinal curvature through muscle recruitment. **Conclusion:** This parametric, fast-solving, high-fidelity spine simulation platform could be a useful biomechanical tool for spine researchers. **Significance:** A novel human torso model with integrated rigid and flexible bodies was presented in this study, providing insights into mechanical spine stability.

Index Terms— lumbar spine, biomechanics, flexible bodies, muscle recruitment, intra-abdominal pressure, thoracolumbar fascia

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I. INTRODUCTION

Computational biomechanics has emerged as an indispensable tool in modern orthopaedic research, enabling detailed investigations of musculoskeletal function, injury mechanisms, and surgical outcomes through *in silico* simulations. Specific to spine, these simulations have proven particularly valuable to understand disease conditions such as low back pain (LBP), which is a leading cause of global disability [1]. Diagnosing the exact cause of LBP remains challenging as it is often presented with non-specific symptoms, likely originating from a combination of anatomical, physiological, biomechanical, and psychological factors. From a purely biomechanical perspective, researchers have associated pain with muscle dysfunction, where the spinal stabilizing muscle recruitment pattern is suboptimal [2], [3]. This could in turn lead to spinal instability and pain.

In vivo investigations into these muscle recruitments, or muscle forces, and intervertebral disc (IVD) stress and strain fields are limited and often impossible to conduct ethically. *In silico* modeling techniques are well suited to address this challenge non-invasively, offering critical insights into the underlying pathomechanisms and potential therapeutic interventions.

Three major types of *in silico* spine models are used in literature: rigid body dynamic (RBD) models (also called multibody models) [4], [5], [6], [7], [8], [9], [10], [11], finite element (FE) models [12], [13], [14], [15], [16], [17], [18], and integrated rigid-flexible body dynamic (RFBD) [19], [20], [21], [22], [23], [24], [25], [26], [27] models. RBD models treat all the skeletal bodies as rigid components connected using joints and constraints to predict the model kinematics and kinetics. Due to their simplified architecture, they are computationally efficient and find uses in robotic applications as digital twins. On the other hand, FE models discretize all the anatomical structures into smaller elements and offer comprehensive geometry-wide stress and strain solutions for a given set of boundary conditions. Due to this, they are heavily dependent on the quality of input tissue geometry and are computationally demanding. Despite this, they are favoured for biomechanical modeling by researchers due to their physics-based approach offering transparency and

generalizability, unlike data-driven black-box approaches such as machine learning.

A third, hybrid modeling technique combining the RBD and FE methodologies was also explored in the past, where the skeletal bones are represented as rigid bodies, and the soft tissues as deformable flexible bodies with volumetric meshing. Some studies have implemented this hybrid approach by running two independent parallel RBD and FE simulations, with data sharing between the models at each

time step [23], [25], [26], [27]. However, these studies also reported the challenges of both the models converging at a single solution i.e. there is a mismatch of final spinal kinematics between the models for the same input boundary conditions. To address this limitation, Remus et al. introduced an integrated RFBD model of the lumbar spine in ArtiSynth, a java-based physics solver that incorporates both the RBD and FE capabilities [20], [21].

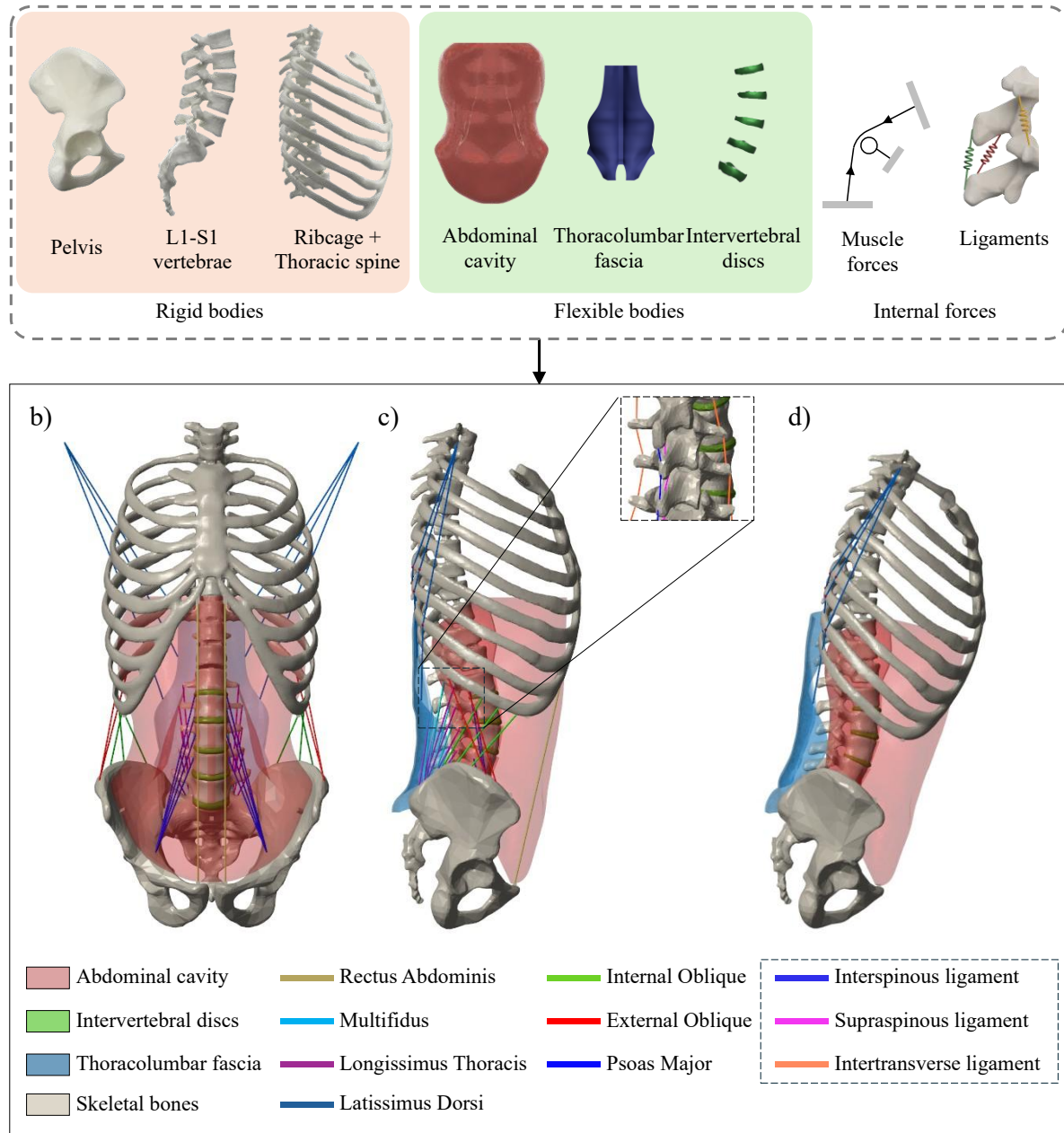


Fig. 1. Schematic of the proposed combined rigid-flexible body dynamic model of the torso. a) Different components included in the model b) Coronal view c) Sagittal view with a detailed view highlighting the three ligament groups in the model d) Example figure highlighting the flexible body deformations in the model under external flexion load.

Building on these advancements, this study presents a MATLAB-native parametric RFBD model of the torso with active and passive musculature, and intra abdominal pressure (IAP) contributing towards spinal stability, while retaining the

advantage of lower computational demand (The MathWorks Inc., MA, USA) [11]. A lower simulation time was a key study objective in order to perform muscle recruitment optimization simulations, and tissue-specific spinal stability

analysis after model validation.

Spinal stability has been hypothesized in the past to be dependent on three sub systems [28], [29] : 1) intrinsic stability of the passive spinal column, 2) dynamic stability provided by the spinal stabilizer muscles and, 3) the effectiveness of the central nervous system in coordinating these stabilizer muscle forces over time. To assess these three subsystems, an effective spine model would have a validated passive spine, the stabilizer muscle force vectors, and the time-varying recruitment of these forces to stabilize the spine against external loads. Furthermore, IAP has been shown to have an unloading effect on the spine under flexion [30], [31], [32], [33], [34], and the TLF with its possible stress shielding effect [14]. The model presented in this study aims to integrate these key contributions to provide a more comprehensive understanding of spinal stability.

Building upon these foundations, this study aims to extend the current state of the art in spine modeling by developing an efficient and physiologically relevant RFBD framework, with a specific focus on:

1. Developing an integrated parametric RFBD model of the passive ligamentous lumbar spine and validate its range of motion (ROM) under loads up to 7.5 Nm (Figure 1).
2. Developing and integrating spinal stabilizer muscles with activation strategy for control, as well as deformable abdominal cavity and posterior thoracolumbar fascia (TLF) models.
3. Assessing the contributions of these supporting tissues to spinal mechanical stability.

II. METHODS AND MATERIALS

A. Skeletal Components

The model consists of eight rigid bodies: the pelvis, the sacrum, L1 to L5 vertebrae, and a fused thoracic spine and ribcage. These models were sourced from an open-source anatomography database based on a 22-year-old healthy adult male [35]. The subject was 172.8 cm tall, weighed 65.0 kg [36]. The model lordotic angle was measured to be 44.76° , aligning closely with the reported average lordotic angle of $47 \pm 10.07^\circ$ observed in asymptotic younger population [37]. The density of all the rigid bodies is set to 181.59 mg/cm^3 , as reported as the average vertebral bone density among adult population [38].

B. Flexible Bodies

The model consisted of seven flexible bodies: the abdominal cavity, the posterior TLF and the L1 to L5 IVDs. The flexible body formulation in this study adopted the “finite-element import method” described in [39] which implements the Craig-Bampton finite-element model reduction method first introduced in 1968 [40]. This method approximates the flexible body behavior by separating its total deformation into two components: the rigid-body motion of its interface nodes (nodes that interact with other bodies), and the internal deformation derived from its FE model. The internal dynamics are typically reduced to a few dominant modes (first 10 in this study), resulting in compact mass and stiffness

matrices that retain the essential dynamics of the flexible body. Derivation of these matrices was done by utilizing the flexible body model builder in MATLAB Simscape. A 3 mm mesh was chosen for all the bodies along with pseudo-incompressible linear elastic properties ($\nu=0.495$). This mesh size was chosen to balance model fidelity and computational efficiency. While no formal convergence study was conducted, preliminary tests confirmed that the 3 mm size provided a stable solution with acceptable simulation time. The elastic modulus of the flexible bodies was defined as a tunable parameter to allow for model calibration. In the case of the IVDs, the interface frames (faces) are the top and bottom endplates, allowing connection to the adjacent vertebrae. For the abdominal cavity, the interfaces correspond to the diaphragm and the pelvic floor, while for the TLF, they are defined at the iliac crest and the tendinous attachments of the latissimus dorsi. These faces are bonded to their adjacent rigid skeletal structures and musculature, enabling seamless force and moment transmission across the rigid-flexible boundaries.

1) Intervertebral Discs

The intervertebral disc (IVD) elastic moduli (E_{IVD}) were calibrated through inverse parameter identification to match the segmental moment-rotation curves reported in literature within a 1° margin of error. For this, a gradient descent optimization routine was used to minimize the root-mean-square error between the model response and the target *ex vivo* curves in flexion as shown in Figure 2. The regression analysis of literature reported *ex vivo* spine studies conducted by Zhang et al. described that the spinal moment-rotation curves could be described as

$$M = C_3 r^3 + C_2 r^2 + C_1 r + C_F F r \quad (1)$$

where M is the moment in Nm, r is the segmental rotation in degrees, F is the follower compression load in newtons and C_1, C_2, C_3, C_F being constants [41]. These polynomial fits served as the target curves for the optimization. The IVD elastic moduli solutions were derived for follower loads up to 1500 N for validation and the solution set for zero follower load was used in further comparisons. A proportional damping constant of $0.05 \text{ Nm}/(\text{deg}/\text{sec})$ was set for all the IVDs for model stability during simulation.

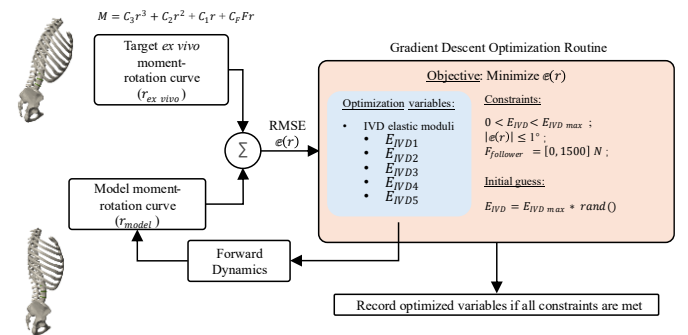


Fig. 2. Optimization routine used to determine IVD elastic moduli.

2) Abdominal Cavity

The abdominal cavity represented the space between the abdominal floor and the diaphragm, offering resistance to

flexion movements through modulation of IAP in humans. The abdominal cavity elastic modulus was used to model the resistance offered by the increase of IAP in the cavity. Cholewicki et al. conducted a series of *in vivo* experiments in 1999 and determined that the trunk stiffness in flexion among ten subjects increased 21% and 42% due to 40% and 80% IAP levels respectively [42]. Based on this, a linear trend was assumed to extrapolate a 53% increase at 100% IAP or normal IAP levels. While this linear relationship cannot be directly validated from the original data, it provides a reasonable approximation in the absence of measurements at full IAP activation. These data points were then used to set the abdominal cavity elasticity. Here the trunk stiffness was defined as

$$K_{trunk} = M_{flexion}/ROM \quad (2)$$

where $M_{flexion}$ is the external moment load on the spine linearly increasing with time, and ROM is the L1-S1 ROM. Once the IVD stiffnesses were tuned, a 7.5 Nm flexion moment at L1 was applied and the trunk ROM was recorded. Then the abdominal cavity was added, and its elastic modulus was manually tuned so that the trunk stiffness increased by 53%, representing normal IAP in adults. A proportional damping constant of 0.1 Nm/(deg/sec) was set for the flexible cavity to reject unwanted oscillations during simulation.

3) Thoracolumbar Fascia

The posterior layer of the thoracolumbar fascia (TLF) was manually segmented from MRI images of a healthy adult male obtained from the Visible Human Project [43], [44]. A different dataset had to be used as the original BodyParts3D dataset did not contain thin walled tissues such as the TLF. The segmented model, approximately 3mm thick, was then scaled to fit the lumbar spine of the RFB model and attachment points were defined on the spinous processes of the vertebrae and on the iliac arch of the pelvis to attach the TLF. The elastic modulus of the TLF was set at 18.8 MPa, an average about the cranio-caudal and latero-medial axes for the thoracic superficial fascia as reported in [45]. The latissimus dorsi were also attached to the TLF and a minimal tension of 18 N [46] was maintained in them to stabilize the tissue during all the simulations. A proportional damping constant of 0.05 Nm/(deg/sec) was used for the TLF body.

C. Spinal Ligaments

Spinal ligaments are collagen rich fibrous soft tissues that provide stiffness to the spine at its physiological end ROM. The major ligaments including the interspinous ligament (ISL), supraspinous ligament (SSL) and the intertransverse ligament (ITL) were modeled as non-linear tension-only springs in this model. The nonlinear force-deformation response of these ligaments was calibrated using the *ex vivo* tensile stress-strain curves reported by Eberlein et al. in [47]. Additionally, the functional cross-sectional areas of these ligaments used to scale the spring stiffnesses were adapted from *ex vivo* readings reported by Shirazi-Adl et al. [48].

D. Musculature

The model consisted of a total of 60 individual muscle fascicles representing seven muscle groups on each side,

namely the multifidus (MF), longissimus thoracis (LT), latissimus dorsi (LD), internal (IO) and external obliques (EO), psoas major (PM), and the rectus abdominus (RA). All these individual muscle fascicles were modeled as point-to-point tensile-only forces between the attachment points on the skeletal rigid bodies. The 3D muscle attachment points were derived from previous radiographic studies and scaled appropriately to fit the current model [49], [50]. Wrapping points were introduced on the ribcage in case of LD to facilitate muscle curvature and force transfer resulting from LD tension and TLF deformation. The physiological cross sectional areas (PCSA) of these muscles were adapted from Cholewicki et al. [51]. The muscle forces, for all muscle groups, were constrained such that the maximum tensile stress in the muscles does not exceed 460 kPa [52].

These muscle groups were then used to explore spinal stability and tissue level contributions to spine stiffness and load sharing during flexion movements in the model.

1) Muscle Recruitment Strategy

The notion of mechanical stability of the spine in this study was inspired by the definition put forth by the Scoliosis Research Society, describing that the spine is said to be stable if the plumbline or the gravity line drawn from the center of C7-T1 IVD falls within ± 2 cm from the posterior superior aspect of the S1 vertebra [53]. This definition only described the stability criterion in the antero-posterior direction which could lead to muscle recruitment solutions that deviate from bilateral symmetry even for symmetric loads. Therefore, an additional constraint along the latero-medial direction was added to obey this bilateral symmetry and to shrink the solution space. The new definition, as also depicted in Figure 3, is as follows:

“The spine is said to be mechanically stable when the plumbline (gravity line) drawn from the center of C7-T1 IVD falls within ± 2 cm from its original position at no-load condition along the latero-medial (X) and the antero-posterior (Y) axes.”

The maximum deviation from the no-load position can then be defined as a measure of mechanical spinal instability U :

$$U = \max(|\Delta x|, |\Delta y|) \quad (3)$$

where Δx and Δy are the plumbline position deviations from the no-load scenario in the latero-medial and the dorsi-ventral axes respectively. The lower the U , the more stable the spine model is. This approach adopts the mechanical stability govern by static equilibrium within an accepted positional tolerance and is consistent with the framework previously proposed by El Bojairami et al. for spine stability [34].

When the spine is outside the stable region, the muscles are activated to bring it back into the stable region. Given the redundant nature of the spine's stabilizers, there could be many solutions as to how these muscle groups are recruited in the human body. In this study, an optimal solution set of muscle forces was determined by way of optimization. The optimality of the solution is defined using a cost function, which represents one of efficiency, that was used by many spine researchers in the past [51], [54], [55]. The cost function

explored in this study is equal to the sum of cubes of all the muscle forces [55], [56]. The optimization routine tries to minimize this cost function while it tries to bring the spine back into the stable region.

$$cost = \sum_{n=1}^N \left(\sum_{i=1}^J f_i \right)^3 \quad (4)$$

where f_i is the force applied at the i^{th} muscle fascicle in the n^{th} muscle group. A global optimization algorithm called surrogate optimization in MATLAB was used to find the optimal muscle forces solution set when the cost function is at the global minimum while maintaining U between the ± 2 cm stability bounds [56] (also see Appendix IIA).

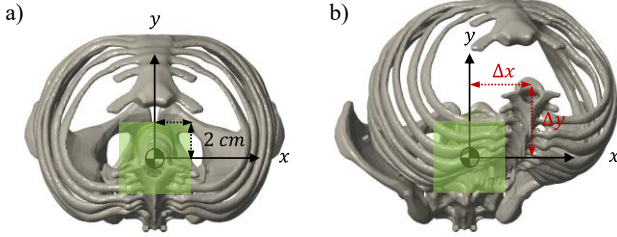


Fig. 3. Mechanical stability of the spine. a) Top view of the undeformed spine. Green square represents the stable region. b) Deformed spine outside the stable region.

E. Contribution of different tissues to spinal stability

After model validation in flexion, a systematic analysis was carried out to quantify the contribution of different tissues to spinal stiffness and load sharing in flexion. Six cases were derived from the model as described below, each focusing on a specific tissue.

- Case 0 (baseline): L1-S1 Only. Only the L1-S1 spine with its ligaments along with the lumped thoracic spine and ribcage were included. This served as a baseline measure to compare against subsequent cases (C0).
- Case 1: L1-S1 + Abdominal cavity. The flexible abdominal cavity was solely included along with the baseline model (C1).
- Case 2: L1-S1 + TLF. The posterior TLF was solely included along with the baseline model. The TLF elastic modulus E_{TLF} was set to 18.8 MPa (C2).
- Case 3: L1-S1 + Muscle forces. Optimized muscle forces to minimize the cost function - sum of cubes of muscle forces (Equation 4), were solely activated along with the baseline model (C3).
- Case 4: L1-S1 + Abdominal cavity + TLF. Both the abdominal cavity (100% IAP) and the TLF ($E_{TLF} = 18.8$ MPa) were included (C4).
- Case 5: L1-S1 + Abdominal cavity + TLF + Muscle forces. The musculature with optimized muscle forces was added to the previous case, completing activation of all the model components (C5).

In each case, the pelvis was fixed, and a 7.5 Nm flexion pure moment was imparted on the models while recording the resulting lumbar rotational displacement and IVD forces for

comparisons. The stability contribution of each case was defined as [34]:

$$Stability\ contribution\ \% = \frac{|ROM_{C0} - ROM_{Ci}|}{ROM_{C0}} \times 100 \quad (5)$$

where, Ci is case 1 to 5; while $C0$ is the baseline (case 0).

F. Model Assembly and Validation Plan

The final model has 30 degrees of freedom (DOFs) distributed across five functional spine units (FSUs), each defined by an adjoining vertebra-IVD-vertebra segment along with their respective ligaments. Each FSU permits six DOFs, three translational and three rotational. Facet joints were not modeled in this version, and no kinematic constraints were imposed at the segmental level, allowing the FSUs to deform freely under load. The lumbar spine is supported by the abdominal cavity, the TLF and the thoracic spine. The model was controlled by internal stabilizing forces comprising of seven muscle groups with a total of 60 muscle fascicles, and three ligament groups with a total of 20 individual ligaments. This system of equations was assembled and compiled in MATLAB R2023a Simscape Multibody (The MathWorks Inc., MA, USA). Subsequently, model verification, validation and analysis followed according to the ASME V&V 40 standard [57].

Model Context of Use:

The model is intended to simulate passive lumbar spine kinematics and kinetics under flexion moments up to 7.5 Nm and compressive follower loads up to 1500 N. Furthermore, it is designed to serve as a general simulation platform to investigate mechanical contributions of different soft tissues to spine stability.

Verification and Validation Framework:

Following the guidelines outlined in ASME V&V 40, a structured framework was designed to verify and validate the model. First, verification activities were carried out to ensure correct representation of the geometry, as well as the RFBD solver consistency (see step 2 below). Then, validation of the model was carried out by comparing model outcome metrics such as segmental and full lumbar ROM, IVD pressure, IAP activation to literature *in vivo*, *ex vivo*, *in silico* and mechanical models.

Analysis:

Post validation, the stability analysis was conducted to estimate the contribution of different soft tissues to spinal mechanical stability. For this, a new definition of mechanical spine stability was proposed, as well as optimization methods to derive optimal muscle recruitment in the model.

Model Risk:

Given the exploratory nature of this model and its use in theoretical research rather than clinical decision-making, the risk associated with this type of model is low-medium.

A step-by-step verification, validation and analysis plan is given below:

- Step 1: First, IVD elastic moduli of the model were set by optimization to match multiple *ex vivo*, *in vivo* and *in silico* literature ranges of motion up to 7.5 Nm flexion load and 1500 N follower load.
- Step 2: Thereafter, to verify the novel flexible body implementation, a similar L1-S1 only finite element model was constructed in ANSYS Workbench 2024 R1 (Canonsburg, PA, USA) with a 3 mm linear tetrahedral mesh, fixed pelvis, rigid vertebrae and optimized IVD elastic moduli. This finite element (FE) model was compared with the RFBD model in ROM up to $M_{flexion} = 7.5$ Nm (2.5 Nm increments) and $F_{follower} = 1500$ N (500 N increments) compressive follower loads, generating a total of 16 load cases. The follower loads were modeled as compressive loads between the adjacent vertebral body centroids and follow the spinal curvature throughout the simulation as shown in Figure 4. Subsequently, the ligaments were removed from each model and tested again, adding 16 additional cases.

ROM and runtime were recorded to benchmark differences between the RFBD and FE implementations. The FE model used the same IVD elastic moduli as the RFBD model, with vertebrae treated as rigid bodies and all contacts assumed bonded. The 3 mm tetrahedral mesh used in the FE model was chosen for consistency with the RFBD implementation. Although a convergence study was not performed here, the FE model was based on a validated framework by El Bojairami et al., whose mesh sensitivity analysis showed $\leq 8.7\%$ variation in intervertebral rotation with finer meshes, supporting the adequacy of this resolution [12].

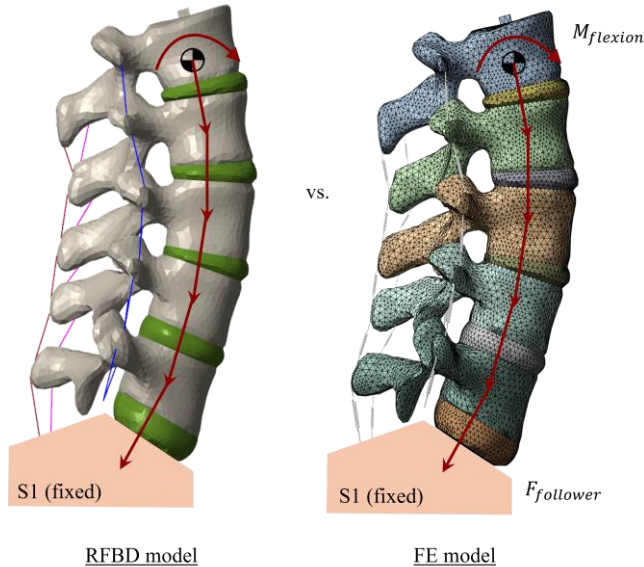


Fig. 4. Boundary conditions for validation of the RFBD model against an identical finite element model.

- Step 3: The abdominal cavity and the TLF elastic moduli were set according to the procedures mentioned in section II B.
- Step 4: Muscle force solutions were determined according to the cost function and constraints mentioned in section II D.
- Step 5: Contribution of different tissues to spinal flexion stiffness and load sharing was analyzed through a systematic comparison as mentioned in section II E.

III. RESULTS

A. IVD Elastic Moduli

The model successfully calibrated its IVD elastic moduli at all variations of follower loads up to 1500 N and 7.5 Nm flexion moment to maintain the required *ex vivo* [41] ROM within $\pm 1^\circ$ error. As the follower load increased from 0 to 1500 N, the average IVD elastic modulus of all five lumbar levels increased linearly from 2.0 ± 0.6 MPa to 13.8 ± 5.6 MPa. Moment rotation curves were plotted for all five segmental levels as well as for L1-S1 with the optimized elastic moduli for zero follower load case. The ROM at all levels match *ex vivo* [58], [59], [60], finite element [60], [61] and rigid body [11] models in literature. As an example, Figure 5 shows the moment-rotation curves at the zero follower load case in flexion. The lumbar spine reached a maximum of 16.01° at 7.5 Nm, with the five vertebral levels contributing 2.46° , 2.76° , 2.50° , 3.44° , 4.91° respectively. The load sharing among the segmental levels was also observed as the maximum reaction moments recorded in the five levels were 3.38, 3.57, 3.08, 2.91, 3.81 Nm respectively. The model was also calibrated and validated against multiple literature models in extension, lateral bending and axial rotation. These graphs were included in Appendix III D.

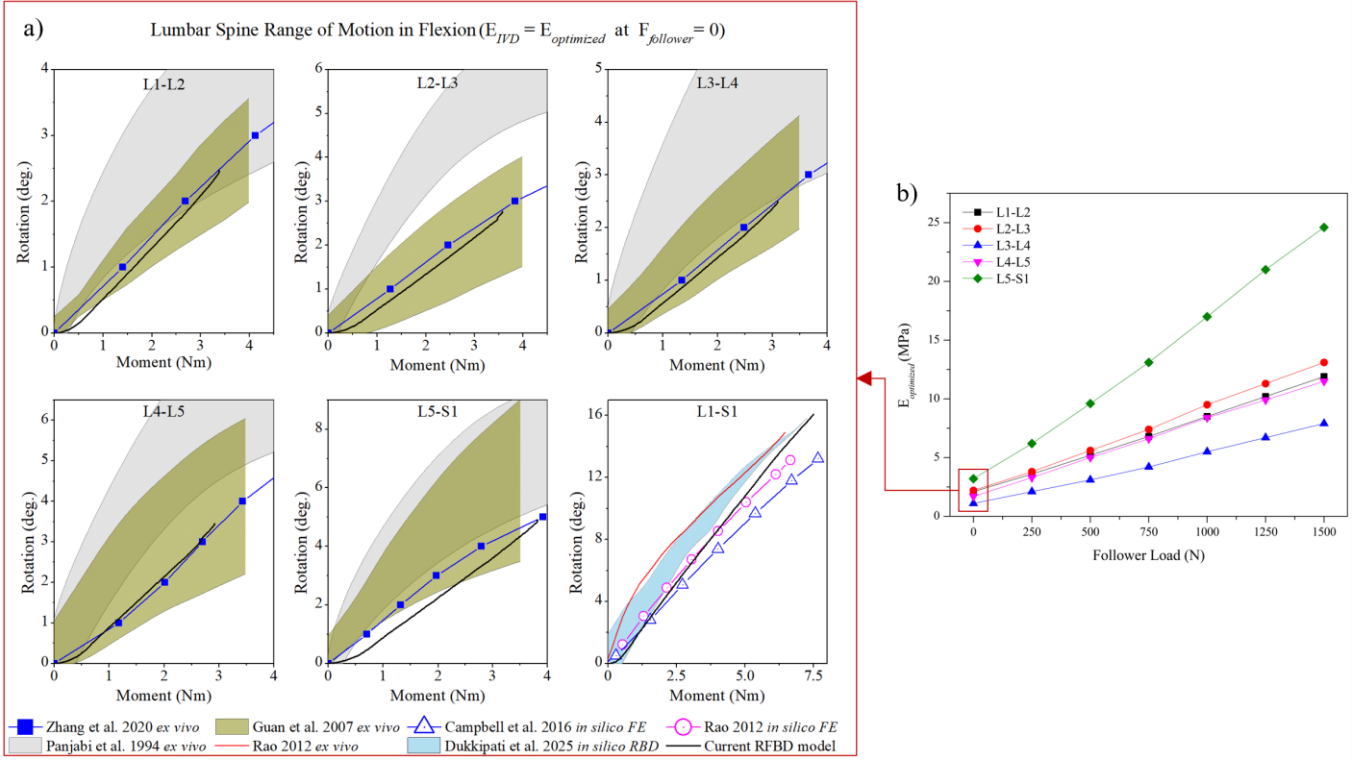


Fig. 5. a) Momen-Rotation curves of the lumbar spine in flexion with optimized IVD elastic moduli at zero follower load case compared to literature *ex vivo* (Zhang et al. 2020 [41], Panjabi et al. 1994 [58], Guan et al. 2007 [59], Rao 2012 [60]), *in silico* finite element (Rao 2012 [60], Campbell et al. 2016 [61]), *in silico* rigid body (Dukkipati et al. 2025 [11]) models. b) Variation of optimized IVD elastic moduli solutions with follower load.

B. Model Validation Against FE Solver

A total of 30 load cases were simulated for both the RFB and FE models, and the L1-S1 ROM and model run time were measured in each. The RFB model ROM compared very closely with the FE model in 27 of 30 cases, as shown in Figure 6. Without ligaments, the maximum ROM variation between the two models was 0.88° . The RFB model had an average runtime of 8.1 ± 0.7 sec, while the FE model required 9.2 ± 0.2 sec. When ligaments were included, the maximum ROM variation between the models was 2.69° . The RFB model runtime remained similar (9.8 ± 0.8 sec),

while the FE model runtime nearly doubled to 20.3 ± 1.5 sec. Furthermore, L4-L5 intradiscal pressure in the model increased linearly, recording 0.85 MPa at 1000 N compressive load, comparing closely with literature *ex vivo* (0.95 MPa average) [62] and *in silico* (0.8 MPa average) [13] models (see Appendix III A).

C. IAP Activation

The IAP activation was achieved by calibrating the abdominal cavity stiffness matrix using its elastic modulus E_{Abd} (see II B and appendix II B).

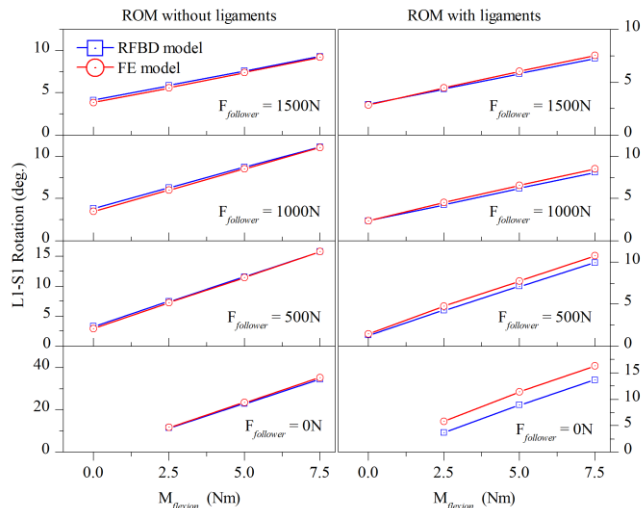


Fig. 6. ROM of the RFB model with and without ligaments compared to identical finite element models.

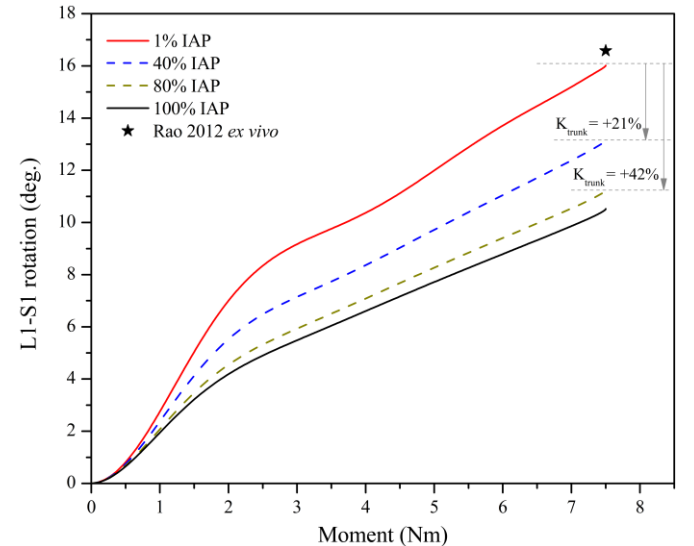


Fig. 7. Lumbar spine range of motion variation at different levels of IAP activation.

Figure 7 shows the decrease in ROM of the ligamentous lumbar spine as the IAP activation increased from 1% to 100%. As prescribed by Cholewicki et al., the model was calibrated to increase the trunk stiffness by 21% and 42% at 40% and 80% IAP activation respectively [42].

The magnitude of compressive and shear forces in all the lumbar segmental levels was also recorded. As IAP activation increased, the average compressive and shear forces decreased as shown in Figure 8. At approximately 200% activation and 7.5 Nm flexion load, the average compressive force in the IVDs reached zero, indicating a stiffer abdominal cavity acts as a lever against the flexion loads, alleviating the compressive load on the spinal column under these conditions. Segmental compressive and shear force distribution along with their directions was included in Appendix III B.

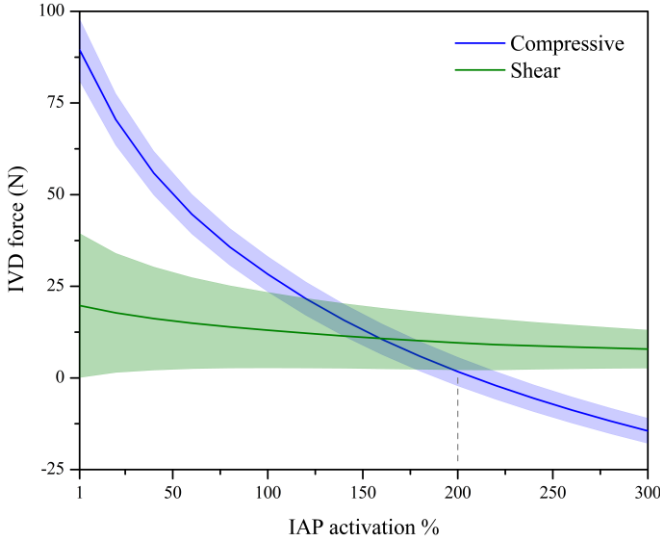


Fig. 8. Variation of IVD compressive and absolute shear forces with increase in IAP activation at 7.5 Nm flexion moment.

D. Muscle Recruitment Optimization

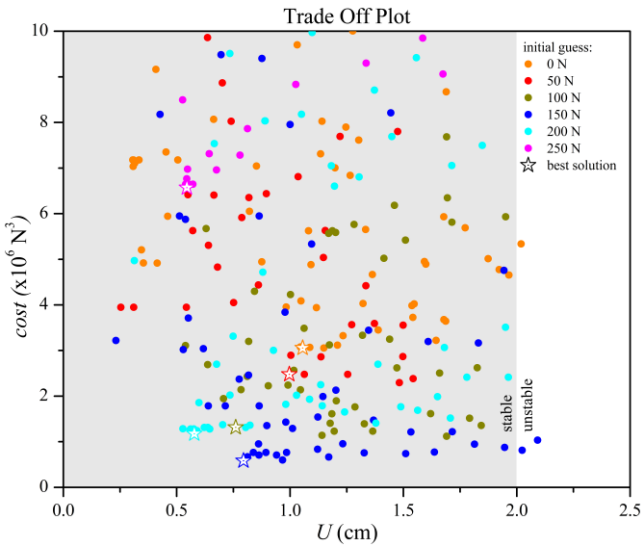


Fig. 9. Trade off plot showing the acceptable range of solutions found by the optimizer for various initial guesses. The best solution from each solution set is marked with a star.

To determine the optimal solution for the muscle forces, the optimization was run using six different initial guesses. The number of maximum iterations for each run was capped at 100 and thus resulted in ~600 optimization iterations. The solution nearest to the origin in each case inside the stable region was designated as the best solution, resulting in six different muscle recruitment patterns that satisfy the imposed constraints.

These six muscle force solutions were used to simulate cases C3 and C5. The trade-off plot for the optimization is shown in Figure 9.

E. Stability Contribution of Different Tissues

Moment-rotation curves for all the cases were plotted up to 7.5 Nm flexion load in Figure 10. The baseline model recorded a ROM of 16.09° as shown previously. Under identical loading conditions, the addition of soft tissues substantially increased trunk stiffness. Addition of the abdominal cavity alone decreased the IVD compressive load by 67% and shear forces by 15%. In contrast, the inclusion of TLF increased both compressive and shear forces by 60% and 30% respectively. Activation of muscles also resulted in a substantial increase in IVD forces as evidenced in cases 3 and 5 as evidenced in Figures 11a and 11b. Out of the three contributors tested, muscle activation proved to be the biggest contributor to stability at 77%. When all the soft tissue contributions were superimposed, the spine stability was estimated to be 81.41%. These results are reported in Table 1. As a reference, the ROM in the baseline model without ligaments was 35.4°, highlighting the potential for spinal instability following ligament injury. Their isolated contribution was not included in the table, as it could not be directly compared to El Bojairami et al., who did not report ligament-only values.

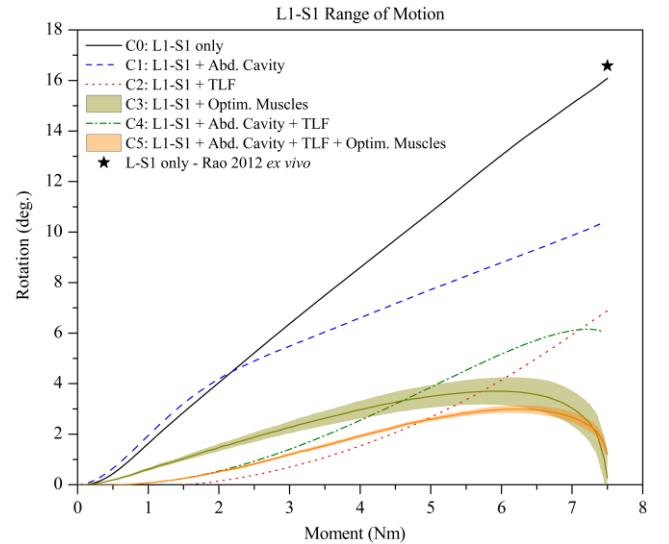


Fig. 10. Lumbar spine range of motion under 7.5 Nm flexion load in all cases C0 to C5. Shaded area represents one standard deviation.

TABLE I
STABILITY CONTRIBUTION OF DIFFERENT CASES

Case	Soft tissues included	Max ROM (deg.)	Trunk Stiffness (Nm/deg)	Stability Contribution %	Stability Contribution % from El Bojairami et al. [34]
0	-	16.09	0.47	-	-
1	Abd. cavity	10.52	0.71	34.61	25
2	TLF	6.89	1.09	57.19	75
3	Muscle forces	3.70	2.03	77.00	53
4	Abd. cavity, TLF	6.16	1.22	61.73	78
5	all	2.99	2.51	81.41	93

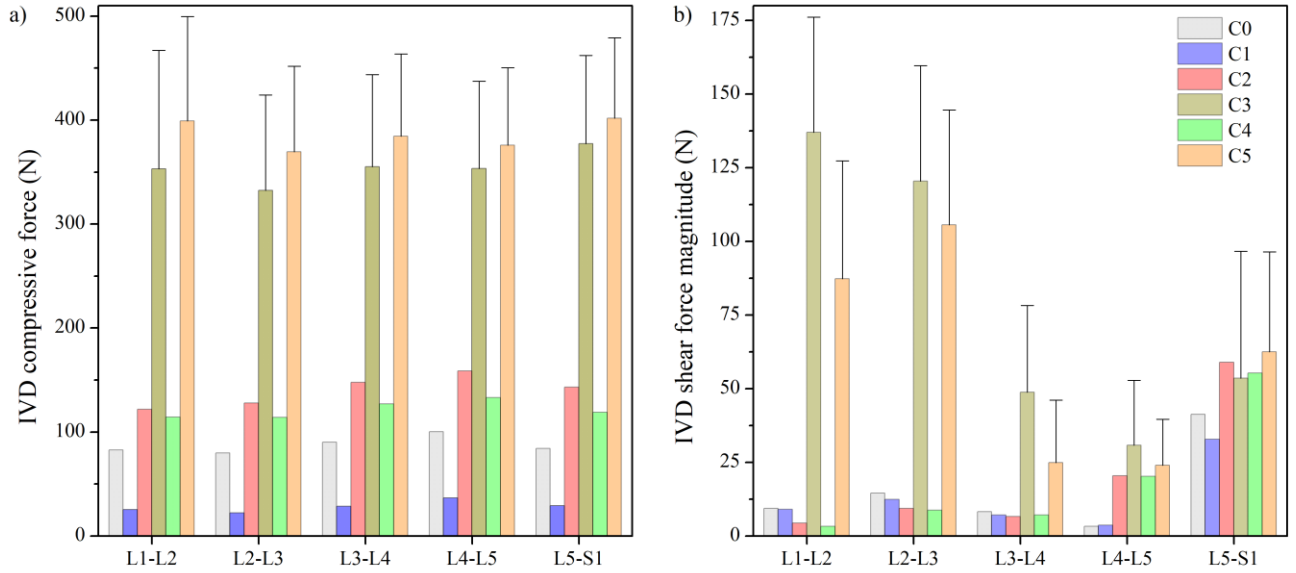


Fig. 11. IVD compressive (a) and shear (b) forces in all the cases C0 to C5. Error bars indicate one standard deviation

F. Model Simulation Time

To benchmark the computational efficiency of this reduced order flexible body modeling approach, the compile and run times in all cases were measured for $n=5$ consecutive runs and presented in Table 2. A 7.5 Nm flexion load at L1 was applied and the models were simulated for one second. All the simulations were performed on a 2.3 GHz, 64-bit laptop with 16 Gb of memory. As expected, case 5 with all the anatomical components included, had the longest average total simulation time of 7.2 minutes, while the simpler baseline model only took an average of 3.33 sec. This is notably on par with the previous RBD version of this model (2.83 sec) [11], and remains competitive with similar hybrid rigid-flexible approaches in literature. For instance, Remus et al. presented a rigid-flexible ligamentous L1-S1 spine model using the ArtiSynth simulation platform achieving an average simulation time of 11 sec [20]. It is to be noted that these benchmarks heavily depend on the computer specifications the model is simulated on, the boundary conditions, and the simulation time etc., and thus cannot be compared directly with models implemented in other softwares and systems. The current RFB model implementation runs on a single CPU core; however, MATLAB also supports parallelization via its parallel computing toolbox, which can be leveraged to

accelerate simulations on multi-core machines. Additionally, GPU-based simulation, also supported by MATLAB, presents a promising avenue for further reducing model run time and is to be explored in future developments. As reported, the proposed approach will be faster than the standard finite element implementation of similar geometric complexity which may often take multiple days to optimize [55]. For ease of comparison with future models, the benchmarking code along with all the case-wise models are made available in the online repository (see Appendix).

TABLE II
MODEL AVERAGE SIMULATION TIME IN SECONDS

Case	Compile time	Run time
0	1.15	2.18
1	1.17	55.09
2	1.20	109.18
3	1.15	2.25
4	1.29	259.10
5	2.14	431.52
RBD model [11]	1.13	1.70

IV. DISCUSSION

A parametric rigid body model of the spine and torso was previously developed as the digital twin of a physical robotic

spine simulator currently being developed [11]. This robotic platform was conceived as a hands-on simulation tool to intuitively visualize the effects of external (e.g., attaching an weight such as a backpack to the robot) and internal (e.g., different muscle recruitment patterns via selectively activating the pneumatic artificial muscles on the robot) forces on the spine in real time, and to potentially be used as a medical device development tool (MDDT). As a part of this effort, two ways of modeling IAP in the context of rigid body models with an emphasis on computational efficiency were proposed in the RBD version of this model while validating one of them against the *in vivo* readings. While this model had a lot of benefits, it also had some limitations. The IVD joints were modeled as nonlinear torque-driven 3 DOF joints ignoring the vertebral translations and corresponding deformations. The musculature was also limited to the available physical muscle groups on the robot. While these assumptions do not affect that study's objective, i.e. developing a digital twin of the spine simulator robot, it limited its applications. This research is a continuation of that work and aims to address all the aforementioned limitations.

The IVDs, the abdominal cavity and the TLF in this study were modeled as 6 DOF deformable flexible bodies, while benefiting from the lightweight rigid body modeling architecture. These components seamlessly integrate with the surrounding rigid bodies using their interface nodes, allowing both rigid and flexible bodies to be modeled within the same framework without requiring additional coding. This approach preserves computational efficiency while enabling the incorporation of localized deformations where needed. Furthermore, the flexible bodies were also coupled bidirectionally with the corresponding adjacent rigid bodies to facilitate load transfer back and forth in the same simulation, as opposed to some uncoupled models proposed in literature. This was particularly helpful in the case of IVDs, as previous studies reported mismatches in kinematics between their rigid and deformable bodies as a result of employing one way coupling between the vertebrae and the IVDs [63], [64], [65]. Lumbar spine stiffness, when submitted to moments up to 7.5 Nm, showed good correlation with similar *ex vivo* and *in silico* models in literature. Load sharing between the vertebrae was also evident in the peak reaction moments as a result of the aforementioned two-way coupling.

The proposed RFBD approach emphasizes computational efficiency through the use of reduced-order flexible bodies, resulting in faster simulations (2.18 sec versus 11 sec in Remus et al. [20]) - a key advantage for iterative applications such as optimization and reinforcement learning. In contrast, the model by Remus et al. offers detailed nonlinear representations of passive spinal structures (e.g., ligamentous tissues, disc fibers) across multiple loading scenarios, albeit at the cost of increased complexity and computational demand. To maintain a manageable model size and simulation time, only key muscle groups relevant to flexion and major ligaments were included in the present study. Although tissues such as the anterior/posterior longitudinal ligaments, and ligamentum flavum were omitted, their mechanical

contributions during flexion are implicitly captured in the optimized IVD stiffnesses. Future iterations of the RFBD model will aim to further explore and optimize the balance between biomechanical fidelity and computational efficiency.

The inclusion of a fully flexible abdominal cavity enabled the model to mimic the apparent stiffness offered by the IAP in flexion movements typically seen in *in vivo* studies. As the IAP activation increased, at a constant flexion load, the IVD compressive forces underwent an exponential decrease, eventually reaching near zero loads around 200% activation. While this fictitious specific datapoint could not be compared directly to any literature, many studies in the past reported this spinal unloading phenomenon as a direct result of IAP. For example, Liu et al. reported a 31% reduction in IVD compressive forces and 24% reduction in shear forces with the inclusion of 26.6 kPa IAP in their model [25]. Arjmand et al. similarly observed a 19% decrease in IVD compression at 4 kPa IAP in standing and 9 kPa IAP flexion poses [66]. Park et al. also reported an average reduction of joint forces from 0.8 to 0.3 times body weight as IAP increased from 0 to 10 kPa [67]. Complementing these modeling studies, Hodges et al. through a series of *in vivo* experiments, found that an increase in 27-61% of IAP resulted in an increase in trunk flexion of 8-31% [30]. This matches the observations made by Cholewicki et al. in their IAP experiments which were used to model IAP activation in this study (also see Appendix II B) [42]. The IAP model used in this study also showed a strong agreement with the previously proposed RBD modeling friendly spring-damper based approach [11]. As these models are parametric, they can be calibrated to match any updated comparator data from future studies.

In the past 50 years, many researchers hypothesized and defined spinal stability but there has not been a clear consensus in the spine biomechanics community [29], [53], [68], [69], [70], [71], [72], [73]. This study adopted a simpler geometry based mechanical stability criterion for the spine to optimize muscle recruitment. Depending on the choice of stability criterion, the muscle recruitment patterns would differ. This study also adopted the widely used cost function (Equation 4) that minimizes muscle effort during recruitment. Some other cost functions, or activation strategies, include minimizing total muscle stress [74], minimizing IVD forces [75], maintaining quasi static equilibrium at all costs [52], minimizing or reversing deformity in a scoliotic spine [76] and minimizing or maximizing intramuscular pressure levels [55] to name a few. The choice of these constraints directly affects the outcome of the optimized muscle force solution set. Therefore, care must be taken when comparing the results of this study to future research. The authors also realized that the choice of the optimization algorithm directly affected the final results. Previous RBD version of this model used a random-initialized gradient descent algorithm, which converged the solution in a local minimum nearest to the initial guess [11]. In contrast, this study used a global optimizer called surrogate optimization [56] which, although computationally expensive, is better suited for applications like muscle recruitment. This is because the underlying cost function of the recruitment could be highly nonlinear and non-smooth, making local

optimization methods prone to converge at suboptimal local minima closest to the initial guess. Surrogate optimization, on the other hand, explores the global landscape of the cost function and converges at the global minima irrespective of the initial guess. As a result, while it takes longer to converge, it yields a more robust and physiologically plausible solution. Ideally, resource permitting, if the optimization were to continue beyond 100 iterations, all the best solutions would eventually converge on a single solution set at the global minima in Figure 9.

The case-by-case analysis of stability contribution including individual tissues and their combinations were explored under an external moment. Under a 7.5 Nm pure moment load, case 0 provided a credible reference for all the subsequent cases in spinal stiffness and IVP pressures. When the abdominal cavity was included (case 1), it lowered the lumbar ROM from 16.09° to 10.52°, contributing about 35% to spinal stability as defined. Notably, the IVD compressive forces were significantly reduced, agreeing with its current understanding of the spinal unloading mechanism. When the TLF was included alone, it acted as a stiff posterior ligament on the spine, contributing about 57% to stability. When the muscles were recruited alone, as expected, agreeing with the implemented recruitment policy, they contributed about 77% to spinal stability. When all the tissues were included, the model achieved the greatest stability at 81% (Table 1). This trend of increased stiffness and the IVD pressures with the progressive inclusion of tissues match well with a similar study by El Bojairami et al. [34]. However, some methodological differences make a direct comparison between these two studies challenging. El Bojairami et al. quantified spinal stability using vertebral translations, whereas our study adopted the plumbline method defined in section II D. Additionally, the boundary conditions of the analyses differed significantly. They analyzed the spine under a horizontal flexion force of 350 N at T1, while this study analyzed it under 7.5 Nm flexion moment at L1. Their muscle forces were derived from EMG-informed inputs reported in the literature, while our model used muscle activations obtained from optimization. Despite these distinctions, both studies observed an overall increase in spinal stability with the addition of spinal stabilizers, highlighting the physiological relevance of these tissues in enhancing spinal stability. Notably, while their study is a finite element implementation, using the proposed integrated rigid-flexible modeling architecture significantly decreased simulation time from hours to minutes per run (Table 2).

An evident application of such a model presented herein is in the investigation of muscle recruitment in healthy and diseased patients. Throughout the literature, many spine researchers have eluded that a cause of certain types of low back pain could be due to muscle dysfunction, i.e. improper or irregular recruitment of muscles to stabilize the spine under external loading. Given the inherent over actuated nature of the spine and its redundant muscle actuators, this could be modeled as an optimization problem, like the methods described in this study, or a reinforcement learning problem. Both methods are iterative and would require multiple

simulations to converge on a solution, for which it is necessary that the underlying model is computationally lightweight. This study aimed to reduce these simulation times, providing a fast-solving simulation platform for future researchers to use. Another application of this model could be as a digital twin of the physical robotic spine simulator the authors are currently developing. This robotic spine has flexible pneumatic artificial muscles for musculature, pressure controlled abdominal and thoracic cavities for IAP, and a stiffness tuned and validated 3D printed lumbar spine [77], [78]. Using the optimized muscle force outputs from the proposed model in this robotic spine could increase its robustness.

A. Limitations and Future Studies:

The IVDs were defined by a linear isotropic elastic modulus and thus are incapable of displaying any nonlinearity. This is evident from the moment rotation curves in Figure 5, (Figures 8, 9 and 10 in Appendix III D) as well as the linear increase in the optimized IVD elastic moduli as the imposed follower load increased. While this may not be of concern under low ranges of moments, the model is expected to deviate from the *ex vivo* nonlinear stiffness behavior above 20 Nm. Furthermore, the current IVD modeling approach does not explicitly incorporate the annulus fibrosus fibers, which are known to significantly influence the nonlinear and anisotropic mechanical behavior of IVDs.

The IAP in this model was varied using the elastic moduli of the abdominal cavity. While this may be one approach to model IAP in these multi body simulations, in humans, it was shown to be activated by recruiting the transverse abdominus muscles [18]. This recruitment constricts the cavity along its circumference, increasing its hoop stress and thereby stretching the TLF and stabilizing the spine [18], [33]. This phenomenon could not be modeled in the current study for simplicity and is to be explored in future iterations. It is also of interest to model the direction-specific stiffness properties of the TLF through embedded fiber elements in the flexible bodies for a more accurate representation of its anisotropy.

The spinal stability criterion adopted in this study heavily promotes the notion that the spine is stable when it is the most rigid, albeit even if activation was driven by efficiency as well. This may not be true in all cases, especially in children. They are considered to have stable and healthy spines, yet they are objectively more flexible than adults [79], [80]. Future studies will aim to incorporate more stability criteria and verify the resultant muscle recruitment against EMG studies. Moreover, non-zero shear force magnitudes were observed in the case-by-case stability analysis, as presented in Figure 11b. This indicates that the sum of all reaction forces at each vertebral level does not align tangentially with the spinal curvature, contrary to what has been suggested by past researchers through the concept of follower load. This could also be enforced as a constraint in future muscle recruitment policies.

V. CONCLUSION

This study presented parametric, biomechanical, integrated rigid-flexible body model of the spine and torso, along with

supporting ligaments, musculature, abdominal cavity and the TLF to perform fast-solving simulations in flexion movements. First, the model was validated in its ROM against *ex vivo* and *in silico* models. Then the flexible body implementation was validated against the conventional finite element solver. Muscle recruitment was modeled as an optimization problem and solved using a modified definition of spinal mechanical stability from literature.

Thereafter, the model was used to explore spinal stability contribution of its constitutive tissues. Six on-off cases were analyzed to determine the individual and combined tissue contributions to stability. The findings reinforce the notion that the intra-abdominal pressure acts as a spinal unloading mechanism in flexion. The stability analysis provided an understanding of how specific tissues may be selectively adapted to improve torso stability in humans. The high-fidelity model presented here could be an efficient simulation platform for spine biomechanicists, physiotherapists, roboticists and educators.

VI. DATA AVAILABILITY

The models and the appendix are available in the online repository: <https://github.com/siril-teja/ROMmodel-io.git>.

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